

Generative AI for Finance: Embedding Numerical Information into Firm Sequences

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Overview

1. Introduction

2. Design

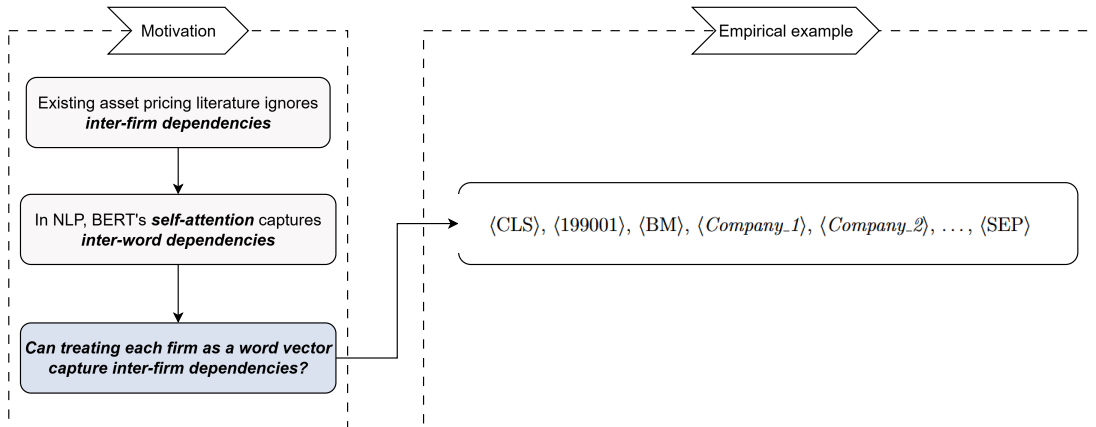
3. Results

4. Replication

5. Ideal

6. Appendix

Summary



Question

1. Can **BERT capture rich firm dependencies** to improve asset pricing performance?
 - 1.1 Lower pricing errors?
 - 1.2 Higher portfolio returns?
2. What is the source of this improvement?
 - The importance of firm rankings

Motivation

- **Firm dependencies** plays a vital role in asset pricing (Kelly et al., 2022, JF)
- However, traditional asset pricing studies **omit inter-firm dependencies** due to a lack of well-established methodologies
 - Like NLP's bag-of-words: treating words as independent entities
- In NLP, **BERT** captures contextual dependencies between word vectors
- This paper: **In cross-sectional pricing, can BERT capture rich inter-firm dependencies by treating characteristic-sorted firms as high-dimensional vectors?**

Marginal Contributions

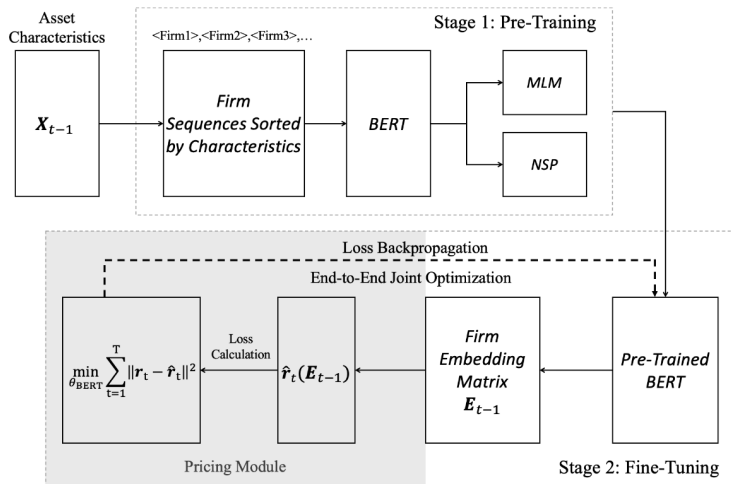
- **Firm dependencies in asset pricing**

- **Prior work:** Overlook inter-firm dependencies (Fama and French, 2015, JFE; Gu et al., 2019, RFS)
- **This paper:** Propose a BERT-based approach to capture complex, high-dimensional inter-firm dependencies

- **GenAI in asset pricing**

- **Prior work:** Use text-extracted metrics and prompt-based methods to **indirectly** link GenAI and asset pricing (Chen et al., 2026, WP; Bie et al., 2025, JFM; Breitung and Müller, 2025, JFE)
- **This paper:** Integrate GenAI **directly** into asset pricing through an end-to-end architecture

N-BERT Model Architecture and Training Workflow



Representative input sequences

- Sequence 1: $\langle \text{CLS} \rangle, \langle 199001 \rangle, \langle \text{BM} \rangle, \langle \textit{Company_1} \rangle, \langle \textit{Company_2} \rangle, \dots, \langle \text{SEP} \rangle$
- Sequence 2: $\langle \text{CLS} \rangle, \langle 199002 \rangle, \langle \text{Size} \rangle, \langle \textit{Company_1} \rangle, \langle \textit{Company_3} \rangle, \dots, \langle \text{SEP} \rangle$

Data

- 94 stock-level predictive characteristics in Green et al. (2017, RFS)
- Monthly stock returns are obtained from CRSP
- January 1980 to December 2023
- Out-of-sample period: January 2001 to December 2023

Q1.1: Can N-BERT achieve lower pricing errors?

Model	R^2_{total}	Relative Pricing Error (%)
PCA	3.36	96.92
LASSO	8.17	90.11
E_Net	9.46	89.14
IPCA5	10.58	84.43
RF	11.26	81.68
XGB	11.80	79.92
AE	13.95	74.77
N-BERT	17.92	66.11

Q1.2: Can N-BERT achieve higher portfolio returns?

- Long-short portfolios returns (High-Low)

Model	SD (%)	Avg (%)	SR
PCA	4.11	0.73	0.18
Lasso	3.12	0.93	0.30
E_Net	3.22	0.96	0.30
IPCA5	3.85	1.39	0.36
RF	3.81	1.48	0.39
XGB	4.18	1.46	0.35
AE	3.53	1.64	0.46
N-BERT5	3.09	2.54	0.82

Q2: What is the source of this improvement?

- S represents shuffled firm sequences
- Firm rankings are important

Model	R^2_{total}	Relative Pricing Error (%)
SN-BERT1	5.31	93.82
SN-BERT2	5.45	96.56
SN-BERT3	4.98	96.13
SN-BERT4	5.10	94.81
SN-BERT5	4.73	93.29
SN-BERT5	4.73	93.29
N-BERT	17.92	66.11

Pricing errors

- Replication: Selected a sample of 1,000 stocks

Model	Original		Replication	
	R^2_{total}	Relative Pricing Error (%)	R^2_{total}	Relative Pricing Error (%)
LASSO	8.17	90.11	2.90	59.96
E_Net	9.46	89.14	3.24	58.19
IPCA	10.58	84.43	8.86	47.53
AE	13.95	74.77	11.64	52.22
N-BERT	17.92	66.11	2.31	64.23

Portfolio returns

- Replication: Selected a sample of 1,000 stocks

Model	Original			Replication		
	SD (%)	Avg (%)	SR	SD (%)	Avg (%)	SR
Lasso	3.12	0.93	0.30	6.07	0.52	0.09
E_Net	3.22	0.96	0.30	6.33	0.51	0.08
IPCA	3.85	1.39	0.36	4.74	0.81	0.17
AE	3.53	1.64	0.46	8.76	1.44	0.16
N-BERT	3.09	2.54	0.82	4.23	0.90	0.21

Extension

1. Inputting sequence
 - Sorting firms based on **upstream and downstream supply chain relationships or other economic linkages**
2. Objective function
 - **Binary classification tasks** (e.g., financial fraud, credit default)
3. Imposing **economic constraints (priors) on different attention heads** to enhance interpretability

Total R^2 (R^2_{total})

The out-of-sample total R^2 is calculated as:

$$R^2_{total} = 1 - \frac{\sum_{i,t \in OOS} (R_{i,t+1} - \hat{R}_{i,t+1})^2}{\sum_{i,t \in OOS} R^2_{i,t+1}}$$

Notation Details:

- $R_{i,t+1}$: The realized excess return of firm i at time $t + 1$.
- $\hat{R}_{i,t+1}$: The predicted return generated by the model for firm i at time $t + 1$ (e.g., constructed by estimated factor loadings and latent factors).
- $(R_{i,t+1} - \hat{R}_{i,t+1})^2$: The squared prediction residual.

Relative Pricing Error

The relative pricing error is calculated as:

$$\text{Relative Pricing Error} = \frac{\sum_{i \in \mathcal{T}} \alpha_i^2}{\sum_{i \in \mathcal{T}} \bar{R}_i^2}$$

where α_i is defined as:

$$\alpha_i = \frac{1}{T} \sum_{t \in \mathcal{T}} (R_{i,t+1} - \hat{R}_{i,t+1})$$

Notation Details:

- $\mathcal{T}$: The set of time periods in the out-of-sample testing window.
- T : The total number of periods in $\mathcal{T}$.
- α_i : The time-series average pricing error (mean prediction residual) for firm i .
- $\bar{R}_i$: The time-series average realized return for firm i over the testing window $\mathcal{T}$.